MSG No. 194.269 $P6'_3/m'mc'$ [Type III, hexagonal]

Table 1: Wyckoff site: `2a`, site symmetry: `-3m.`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`
2	$\left[0, 0, \frac{1}{2}\right]$	`[13,14,15,16,17,18,19,20,21,22,23,24]`

Table 2: Wyckoff site: `2b`, site symmetry: `-6'm2'`

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	`[1,2,3,10,11,12,16,17,18,19,20,21]`
2	$\left[0, 0, \frac{3}{4}\right]$	`[4,5,6,7,8,9,13,14,15,22,23,24]`

Table 3: Wyckoff site: `2c`, site symmetry: `-6'm2'`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{4}\right]$	`[1,2,3,10,11,12,16,17,18,19,20,21]`
2	$\left[\frac{2}{3}, \frac{1}{3}, \frac{3}{4}\right]$	`[4,5,6,7,8,9,13,14,15,22,23,24]`

Table 4: Wyckoff site: `2d`, site symmetry: `-6'm2'`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, \frac{3}{4}\right]$	`[1,2,3,10,11,12,16,17,18,19,20,21]`
2	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{4}\right]$	`[4,5,6,7,8,9,13,14,15,22,23,24]`

Table 5: Wyckoff site: `4e`, site symmetry: `3m.`

No.	position	mapping
1	$[0, 0, z]$	`[1,2,3,10,11,12]`
2	$[0, 0, -z]$	`[4,5,6,7,8,9]`
3	$\left[0, 0, z + \frac{1}{2}\right]$	`[13,14,15,22,23,24]`
4	$\left[0, 0, \frac{1}{2} - z\right]$	`[16,17,18,19,20,21]`

Table 6: Wyckoff site: `4f`, site symmetry: `3m.`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	`[1,2,3,10,11,12]`
2	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	`[4,5,6,7,8,9]`
3	$\left[\frac{2}{3}, \frac{1}{3}, z + \frac{1}{2}\right]$	`[13,14,15,22,23,24]`
4	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2} - z\right]$	`[16,17,18,19,20,21]`

Table 7: Wyckoff site: `6g`, site symmetry: `.2/m.`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	`[1,4,7,10]`
2	$\left[0, \frac{1}{2}, 0\right]$	`[2,5,8,11]`
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,6,9,12]`
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[13,16,19,22]`
5	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[14,17,20,23]`
6	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[15,18,21,24]`

Table 8: Wyckoff site: `6h`, site symmetry: `m'm2'`

No.	position	mapping
1	$\left[x, 2x, \frac{1}{4}\right]$	`[1,10,17,20]`
2	$\left[-2x, -x, \frac{1}{4}\right]$	`[2,11,18,21]`
3	$\left[x, -x, \frac{1}{4}\right]$	`[3,12,16,19]`
4	$\left[-x, -2x, \frac{3}{4}\right]$	`[4,7,14,23]`
5	$\left[2x, x, \frac{3}{4}\right]$	`[5,8,15,24]`
6	$\left[-x, x, \frac{3}{4}\right]$	`[6,9,13,22]`

Table 9: Wyckoff site: `12i`, site symmetry: `.2.`

No.	position	mapping
1	$[x, 0, 0]$	`[1,4]`
2	$[0, x, 0]$	`[2,5]`
3	$[-x, -x, 0]$	`[3,6]`
4	$[-x, 0, 0]$	`[7,10]`
5	$[0, -x, 0]$	`[8,11]`
6	$[x, x, 0]$	`[9,12]`
7	$\left[x, x, \frac{1}{2}\right]$	`[13,16]`
8	$\left[-x, 0, \frac{1}{2}\right]$	`[14,17]`
9	$\left[0, -x, \frac{1}{2}\right]$	`[15,18]`
10	$\left[-x, -x, \frac{1}{2}\right]$	`[19,22]`
11	$\left[x, 0, \frac{1}{2}\right]$	`[20,23]`
12	$\left[0, x, \frac{1}{2}\right]$	`[21,24]`

Table 10: Wyckoff site: `12j`, site symmetry: `m'..`

No.	position	mapping
1	$\left[x, y, \frac{1}{4}\right]$	`[1,20]`
2	$\left[-y, x-y, \frac{1}{4}\right]$	`[2,21]`
3	$\left[-x+y, -x, \frac{1}{4}\right]$	`[3,19]`
4	$\left[x-y, -y, \frac{3}{4}\right]$	`[4,23]`
5	$\left[y, x, \frac{3}{4}\right]$	`[5,24]`
6	$\left[-x, -x+y, \frac{3}{4}\right]$	`[6,22]`
7	$\left[-x, -y, \frac{3}{4}\right]$	`[7,14]`

continued ...

Table 10

No.	position	mapping
8	$\left[y,\ -x+y,\ \frac{3}{4}\right]$	[8,15]
9	$\left[x-y,\ x,\ \frac{3}{4}\right]$	[9,13]
10	$\left[-x+y,\ y,\ \frac{1}{4}\right]$	[10,17]
11	$\left[-y,\ -x,\ \frac{1}{4}\right]$	[11,18]
12	$\left[x,\ x-y,\ \frac{1}{4}\right]$	[12,16]

Table 11: Wyckoff site: 12k, site symmetry: .m.

No.	position	mapping
1	$[x,\ 2x,\ z]$	[1,10]
2	$[-2x,\ -x,\ z]$	[2,11]
3	$[x,\ -x,\ z]$	[3,12]
4	$[-x,\ -2x,\ -z]$	[4,7]
5	$[2x,\ x,\ -z]$	[5,8]
6	$[-x,\ x,\ -z]$	[6,9]
7	$\left[-x,\ x,\ z+\frac{1}{2}\right]$	[13,22]
8	$\left[-x,\ -2x,\ z+\frac{1}{2}\right]$	[14,23]
9	$\left[2x,\ x,\ z+\frac{1}{2}\right]$	[15,24]
10	$\left[x,\ -x,\ \frac{1}{2}-z\right]$	[16,19]
11	$\left[x,\ 2x,\ \frac{1}{2}-z\right]$	[17,20]
12	$\left[-2x,\ -x,\ \frac{1}{2}-z\right]$	[18,21]

Table 12: Wyckoff site: 24l, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[-y,\ x-y,\ z]$	[2]
3	$[-x+y,\ -x,\ z]$	[3]
4	$[x-y,\ -y,\ -z]$	[4]
5	$[y,\ x,\ -z]$	[5]
6	$[-x,\ -x+y,\ -z]$	[6]
7	$[-x,\ -y,\ -z]$	[7]
8	$[y,\ -x+y,\ -z]$	[8]
9	$[x-y,\ x,\ -z]$	[9]
10	$[-x+y,\ y,\ z]$	[10]
11	$[-y,\ -x,\ z]$	[11]
12	$[x,\ x-y,\ z]$	[12]
13	$\left[x-y,\ x,\ z+\frac{1}{2}\right]$	[13]
14	$\left[-x,\ -y,\ z+\frac{1}{2}\right]$	[14]
15	$\left[y,\ -x+y,\ z+\frac{1}{2}\right]$	[15]
16	$\left[x,\ x-y,\ \frac{1}{2}-z\right]$	[16]
17	$\left[-x+y,\ y,\ \frac{1}{2}-z\right]$	[17]
18	$\left[-y,\ -x,\ \frac{1}{2}-z\right]$	[18]
19	$\left[-x+y,\ -x,\ \frac{1}{2}-z\right]$	[19]

continued ...

Table 12

No.	position	mapping
20	$\left[x,\, y,\, \frac{1}{2}-z\right]$	[20]
21	$\left[-y,\, x-y,\, \frac{1}{2}-z\right]$	[21]
22	$\left[-x,\, -x+y,\, z+\frac{1}{2}\right]$	[22]
23	$\left[x-y,\, -y,\, z+\frac{1}{2}\right]$	[23]
24	$\left[y,\, x,\, z+\frac{1}{2}\right]$	[24]